

Methodology 3 — Homotopy in δ

Why the final iterate lands at a vertex

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Draft outline — content to come.

The curvature parameter

δ in the relaxed objective controls how strongly the envelope penalty pushes t toward vertices. Small $\delta \rightarrow$ smooth, easy to optimise. Large $\delta \rightarrow$ strongly concave on the polytope.

Homotopy schedule

Start with small δ , increase geometrically:

$$\delta^{(0)} < \delta^{(1)} < \dots < \delta^{(R)} \geq \delta_{\text{conc}}.$$

Warm-start each FW solve with the previous t .

Theorem 1 — concavity threshold

There exists an explicit δ_{conc} (depending on the design and the model) above which $f_{\delta,\lambda}(t)$ is concave on the cardinality polytope, and **every minimiser is a binary vertex**.

For logistic regression:

$$\delta_{\text{conc}} = \frac{\nu_{\max}}{8n}.$$

For multinomial:

$$\delta_{\text{conc}} = \frac{\nu_{\max}}{4n}.$$

(where $\nu_{\max}$ is the largest eigenvalue of $X_u^\top X_u$.)

The k -sweep

Loop over $k = 1, 2, \dots, q$, warm-starting each k from the previous solution. One sweep produces a full subset path $S_1, S_2, \dots, S_q$.

Placeholder for visual: animation of the loss surface deforming with δ from smooth to spiky-at-corners; gif candidate.

Where to next

See it work — in code, on real data:

- [Linear simulation](#)
- [HD logistic simulation](#)
- [Khan SRBCT](#)
- [Rice GWAS](#)
- [Comparisons](#)

← [Frank-Wolfe](#)